

**Master of Computer Application (M.C.A.) (Two Year Programme) -
2nd Semester (CBEGS) (Batch 2024-26)
(2525)**

Paper : CSL-4099 : Theory of Computation

Time allowed: 3 hrs.

Max. Marks: 70

Note: Attempt five questions in all by selecting at least one question from each section. Each question carries equal marks.

Section A

- What is Chomsky Normal form? Reduce the following grammar to CNF: $S \rightarrow ASA \mid aB$
 $A \rightarrow B \mid S, B \rightarrow b \mid \epsilon$. (4,10)
- (a) Define Grammar. Explain Chomsky hierarchy of Grammar. (6)
(b) Write regular expressions for the following languages (4,4)
 - Strings over $\{0,1\}$ containing at least one 1.
 - Strings ending with 00.

Section B

- Design a transition diagram for a DFA accepting strings generated over $\Sigma = \{0, 1\}$ containing two consecutive ones followed by two consecutive zeros. (14)
- Let G be the grammar $S \rightarrow 0B \mid 1A, A \rightarrow 0 \mid 0S \mid 1AA, B \rightarrow 1 \mid 1S \mid 0BB$. For the string 00110101. Find the leftmost derivation, right most derivation and the derivation tree. (5,5,4)

Section C

- Define Moore and Mealy machine. Construct a Moore Machine equivalent to the Mealy machine given in the table below: (14)

Present State	Next State			
	Input a=0		Input a=1	
	State	Output	State	Output
$\rightarrow q_1$	q_3	0	q_2	0
q_2	q_1	1	q_4	0
q_3	q_2	1	q_1	1
q_4	q_4	1	q_3	0

- Define Pushdown Automata. Design a PDA for the language $L = a^n b^{2n}$ for all $n \geq 1$. (14)

Section D

- What is the difference between recursive language and recursively enumerable language? Explain with example. (14)
- What is Post Correspondence Problem? Show how PCP is used to prove undecidability. Give an instance of solvable PCP and its solution. (4,5,5)
